STAT-437: Applied Statistical Methods
Data Analysis Project: Count Regression Models
Due: Oct 30, 2026 at 10:00PM
Total=75 Marks

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Instructions:

- A written Data Analysis Project report and the corresponding R code, including all required tables and graphics, must be uploaded to Canvas by the deadline. The analysis must be reproducible, and all results must be clearly interpreted in the context of the selected dataset.

Overview

The objective of this project is to apply count regression models to a real dataset with a non-negative integer response variable, such as the number of events, visits, accidents, or disease cases. Students will investigate relationships between the count response and explanatory variables, conduct statistical inference, compare candidate models, assess model adequacy, and interpret the findings in the context of a meaningful research question. Students should consider Poisson regression as the primary model and assess whether overdispersion warrants an alternative approach, such as quasi-Poisson or negative binomial regression.

Required Analysis

1. **Data Description and Exploratory Data Analysis (10 marks)**:
   Describe the dataset, define the count response variable, identify the explanatory variables, and present appropriate descriptive statistics and graphical summaries. Examine the distribution of the response, including its mean and variance.
2. **Count Regression Model (10 marks)**:
   Formulate and fit an appropriate Poisson regression model with a log link. Explain why the model is suitable for the selected count response. Present the model formulation, define the variables, identify reference categories where applicable, and interpret the estimated regression coefficients.
3. **Statistical Inference and Overdispersion (10 marks)**:
   Identify important predictors, interpret estimated coefficients and incidence rate ratios (IRRs) with confidence intervals, and conduct appropriate hypothesis tests. Assess overdispersion using suitable measures, such as the Pearson chi-square statistic divided by the residual degrees of freedom. If overdispersion is present, fit and interpret an appropriate alternative model, such as quasi-Poisson or negative binomial regression.
4. **Model Selection and Assessment (10 marks)**:
   Compare appropriate candidate models using AIC and likelihood-ratio tests for suitable nested models. Justify the final model, assess its adequacy using appropriate diagnostics, and evaluate predictive performance using suitable measures. Explain the limitations of AIC and likelihood-ratio tests when comparing models fitted using different estimation methods.
5. **Discussion and Conclusion (10 marks)**:
   Summarize the main findings, discuss the limitations and practical implications of the analysis, and suggest possible future extensions, such as a Generalized Linear Mixed Model (GLMM) when repeated measurements or clustered observations are relevant.

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Report Presentation, R Code, and Reproducibility(25 marks)

- Report Organization and Presentation (10 marks): Include a suitable title page, abstract, keywords, introduction, and clearly organized sections. Present the report professionally.
- R Code Quality and Completeness (5 marks): Submit complete, well-organized, and clearly documented R code for all statistical analyses.
- Reproducibility and Consistency (5 marks): Ensure that the R code reproduces the reported results and that the code and report are consistent.
- Tables, Graphics, and References (5 marks): Present clearly labelled and appropriately formatted tables and graphics. Cite the dataset source and all references used.

Required Report Structure

The project should contain the following sections:

1. **Title Page**: Include an informative project title, course name, group number, and the names and student IDs of all group members.
2. **Abstract**: Briefly summarize the research question, dataset, statistical methods, main findings, and conclusions. Approximately 80-120 words are recommended.
3. **Keywords**: Provide 3-5 keywords describing the research topic and statistical methods.
4. **Introduction**: Explain the research problem, its importance, the research question, and the rationale for selecting the explanatory variables.
5. **Data Description and Exploratory Data Analysis**: Describe the dataset and variables, and present appropriate descriptive statistics and graphical summaries.
6. **Count Regression Analysis**:Present the model formulation, parameter estimates, incidence rate ratios (IRRs), confidence intervals, hypothesis tests, and predicted counts. Where appropriate, also report predicted probabilities for selected count values or events.
7. **Model Selection and Assessment**: Compare candidate models, justify the final model, and discuss relevant diagnostic findings and predictive performance.
8. **Discussion and Conclusion**: Summarize the findings, discuss limitations and practical implications, and suggest possible future extensions.
9. **References**: Cite the dataset source and all references used.
10. **Appendix (if needed)**: Include supplementary tables, figures, or other relevant material. Submit the complete R code separately.

Datasets

Students may select a suitable dataset from an R package or another reliable source. Potential sources include:

- MASS:Statistical and regression datasets.
- datasets: built-in R datasets.
- faraway: Datasets for applied regression analysis.
- boot: Datasets for statistical analysis.
- AER: Applied econometrics datasets.

The selected dataset must contain a meaningful non-negative integer count response and suitable explanatory variables. Students may not use datasets already covered in the course lecture notes, examples, or homework assignments. The dataset does not need to be overly complex. The main objective is to demonstrate an understanding of count regression models and their appropriate application to real data. Students must cite the source of their selected dataset.

General Requirements

- Use R for all statistical analyses.
- State the research question and explain the rationale for selecting the explanatory variables.
- Submit the complete R code used to produce the reported results, tables, and graphics.
- Interpret statistical results in the context of the research question.
- Examine the count distribution and assess whether the Poisson mean-variance assumption is reasonable.
- Check for overdispersion using appropriate measures and explain its implications for statistical inference.
- When appropriate, fit a quasi-Poisson or negative binomial model and justify the choice.
- Use suitable methods for model selection and assessment; do not rely solely on statistical significance to select the final model.
- When comparing suitable nested Poisson or negative binomial models, use an appropriate likelihood-ratio test. Do not apply the ordinary likelihood-ratio test to quasi-Poisson models, because quasi-likelihood models do not specify a full probability distribution.
- Use AIC only when the candidate models have comparable likelihood-based fits; ordinary AIC is not defined for quasi-Poisson models.
- Check relevant model diagnostics, including residual patterns, influential observations, and model-fitting problems where appropriate.
- Evaluate predictive performance using appropriate measures. If a test set or cross-validation is used, describe the procedure and distinguish out-of-sample performance from in-sample performance.
- Cite all data sources and references
- Ensure that the report and R code are consistent, complete, and reproducible.
- Follow the marking scheme provided.

Note: This project focuses on count regression models. Students are not required to fit a GLMM. A GLMM may be discussed as a possible future extension when repeated measurements or clustered observations are relevant.